EXPERIMENT 8

PROBLEM TITLE

Develop a Simple Text Classification Model Using Bag-of-Words.

OBJECTIVE OF THE PROGRAM

To implement a basic text classification model using the Bag-of-Words representation to classify text data into predefined categories.

DESCRIPTION

The Bag-of-Words (BoW) model is a simple method for representing text data in machine learning. It treats each text as an unordered collection of words, disregarding grammar and word order but maintaining multiplicity. Each word is assigned a unique index in a vocabulary, and the text is represented by a vector with word counts corresponding to these indices. This vectorized form is then used as input to a machine learning model for classification tasks.

ALGORITHM

1. Preprocess the text data by tokenizing, removing stopwords, and stemming/lemmatizing.
2. Convert the text data into Bag-of-Words vectors.
3. Split the data into training and testing sets.
4. Train a classification model (e.g., Naive Bayes) using the training data.
5. Evaluate the model using the test data.

EXPERIMENT 9

PROBLEM TITLE

Develop a Simple Text Classification Model Using TF-IDF Representation.

OBJECTIVE OF THE PROGRAM

To implement a basic text classification model using the Term Frequency-Inverse Document Frequency (TF-IDF) representation for improved text vectorization and classification.

DESCRIPTION

TF-IDF is an advanced version of the Bag-of-Words model, where each word's frequency is weighted by its importance, as determined by how often it appears in the entire corpus. TF-IDF helps to diminish the influence of frequently occurring words that carry less informative value across documents. This representation often leads to better classification performance compared to simple word counts.

ALGORITHM

1. Preprocess the text data by tokenizing, removing stopwords, and stemming/lemmatizing.
2. Convert the text data into TF-IDF vectors.
3. Split the data into training and testing sets.
4. Train a classification model (e.g., SVM) using the training data.
5. Evaluate the model using the test data.

EXPERIMENT 10

PROBLEM TITLE
Study of PROLOG Programming Language and its Functions.

OBJECTIVE OF THE PROGRAM
To explore the basics of the PROLOG programming language and create simple facts and rules, demonstrating its use in knowledge representation and logic programming.

DESCRIPTION
PROLOG (Programming in Logic) is a high-level programming language associated with artificial intelligence and computational linguistics. It is based on formal logic and is particularly well-suited for tasks that involve symbolic reasoning, such as natural language processing, knowledge representation, and expert systems. In PROLOG, problems are expressed in terms of relations, represented as facts and rules, and queries are used to infer answers based on these relationships.

ALGORITHM
1. Define simple facts to represent knowledge.
2. Define rules that infer relationships from facts.
3. Query the system to infer new knowledge based on the facts and rules.